

PURPOSE

farm system, crop production cycle, inputs and outputs. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Agriculture Fundamentals: farm system, crop production cycle, inputs and outputs. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Agriculture Fundamentals?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

monsoon crop calendar, rainfall and risk. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Kharif Season: monsoon crop calendar, rainfall and risk. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

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FARMER / RAG QUESTIONS

- What is Kharif Season?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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PURPOSE

winter crop calendar, irrigation and temperature. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Rabi Season: winter crop calendar, irrigation and temperature. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Rabi Season?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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PURPOSE

summer crops, heat and water management. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Zaid Season: summer crops, heat and water management. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Zaid Season?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Use integrated pest management and current source-verified recommendations.
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PURPOSE

orchard and long-duration crop planning. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Perennial Crops: orchard and long-duration crop planning. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Perennial Crops?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
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PURPOSE

matching crop to climate, soil, water and market. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Crop Selection: matching crop to climate, soil, water and market. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Crop Selection?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
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Growth stage:	Vegetative / Flowering / Fruiting / Maturity
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7. Climate and Crop Suitability

PURPOSE

temperature, rainfall, humidity and crop stages. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Climate and Crop Suitability: temperature, rainfall, humidity and crop stages. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Climate and Crop Suitability?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
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PURPOSE

texture, structure, fertility, drainage and crop suitability. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Soil Types: texture, structure, fertility, drainage and crop suitability. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

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MODERN FARMING CONNECTION

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DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Soil Types?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

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Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
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PURPOSE

pH, EC, organic carbon, NPK and micronutrients. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Soil Testing: pH, EC, organic carbon, NPK and micronutrients. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

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FARMER / RAG QUESTIONS

- What is Soil Testing?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

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Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
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PURPOSE

organic matter, biological activity, compaction and erosion. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Soil Health: organic matter, biological activity, compaction and erosion. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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MODERN FARMING CONNECTION

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DATASET FIELDS

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FARMER / RAG QUESTIONS

- What is Soil Health?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

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Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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Action:	Source-backed management
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PURPOSE

purity, germination, vigor, source and storage. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Seed Quality: purity, germination, vigor, source and storage. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Seed Quality?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
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Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
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PURPOSE

purpose, hygiene and source-verified treatment practices. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Seed Treatment: purpose, hygiene and source-verified treatment practices. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Seed Treatment?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
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IMPORTANT NOTES

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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

seedling hygiene, watering, hardening and transplanting. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Nursery Management: seedling hygiene, watering, hardening and transplanting. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Nursery Management?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

timing, spacing, depth and population. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Sowing and Planting: timing, spacing, depth and population. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Sowing and Planting?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

establishment, vegetative, flowering, filling and maturity. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Crop Growth Stages: establishment, vegetative, flowering, filling and maturity. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Crop Growth Stages?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

crop stage, soil moisture and irrigation scheduling. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Irrigation Management: crop stage, soil moisture and irrigation scheduling. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Irrigation Management?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

waterlogging prevention and recovery. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Drainage Management: waterlogging prevention and recovery. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Drainage Management?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

balanced fertilizer and soil-test-based decisions. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Nutrient Management: balanced fertilizer and soil-test-based decisions. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Nutrient Management?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

zinc, iron, boron and other crop-specific deficiencies. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Micronutrients: zinc, iron, boron and other crop-specific deficiencies. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Micronutrients?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

identification, competition and integrated control. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Weed Management: identification, competition and integrated control. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Weed Management?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

breaking pest/disease cycles and improving soil. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Crop Rotation: breaking pest/disease cycles and improving soil. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Crop Rotation?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

resource use, diversity and risk management. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Intercropping: resource use, diversity and risk management. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Intercropping?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

season, stages, water, pests and diseases. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Rice is mainly associated with Kharif. Broad planting reference: June–July. Disease examples: Blast, bacterial leaf blight, sheath blight. Pest examples: Stem borer, leaf folder, brown planthopper. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Rice outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Rice Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Rice
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

season, stages, irrigation, rusts and aphids. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Wheat is mainly associated with Rabi. Broad planting reference: October–December. Disease examples: Stripe rust, leaf rust, powdery mildew. Pest examples: Aphids, termites. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Wheat outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Wheat Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Wheat
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

season, stages, fall armyworm and disease risks. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Maize is mainly associated with Kharif. Broad planting reference: June–July. Disease examples: Turcicum leaf blight, downy mildew, stalk rot. Pest examples: Fall armyworm, stem borer, shoot fly. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Maize outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Maize Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Maize
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

crop stages, boll protection and sucking pests. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Cotton is mainly associated with Kharif. Broad planting reference: June–July. Disease examples: Bacterial blight, wilt, boll rot. Pest examples: Bollworms, whitefly, jassid, thrips. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Cotton outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Cotton Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Cotton
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

monsoon establishment, diseases and insect monitoring. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Soybean is mainly associated with Kharif. Broad planting reference: June–July. Disease examples: Yellow mosaic, rust, charcoal rot. Pest examples: Girdle beetle, stem fly, tobacco caterpillar. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Soybean outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Soybean Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Soybean
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

pod development, leaf diseases and moisture. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Groundnut is mainly associated with Kharif. Broad planting reference: June–July. Disease examples: Leaf spot, rust, collar rot. Pest examples: Aphids, thrips, leaf miner. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Groundnut outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Groundnut Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Groundnut
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

long duration, wilt, sterility mosaic and pod pests. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Pigeonpea is mainly associated with Kharif. Broad planting reference: June–July. Disease examples: Wilt, sterility mosaic, phytophthora blight. Pest examples: Pod borer, pod fly. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Pigeonpea outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Pigeonpea Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Pigeonpea
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

Rabi crop, wilt and pod borer management. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Chickpea is mainly associated with Rabi. Broad planting reference:

October–November. Disease examples: Fusarium wilt, Ascochyta blight, dry root rot.

Pest examples: Gram pod borer, cutworm, aphid. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Chickpea outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Chickpea Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Chickpea
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

Rabi oilseed, aphid and foliar disease risks. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Mustard is mainly associated with Rabi. Broad planting reference: October–November. Disease examples: Alternaria blight, white rust, downy mildew. Pest examples: Mustard aphid, painted bug. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Mustard outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Mustard Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Mustard
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

seed quality, blight, tuber pests and storage. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Potato is mainly associated with Rabi/cool season. Broad planting reference: Region-specific. Disease examples: Late blight, early blight, bacterial wilt. Pest examples: Aphids, tuber moth, cutworm. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Potato outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Potato Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Potato
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

nursery, flowering, fruiting, virus and borer risks. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Tomato is mainly associated with Multiple. Broad planting reference: Region-specific. Disease examples: Early/late blight, bacterial wilt, leaf curl virus. Pest examples: Fruit borer, whitefly, thrips. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruiting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Tomato outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Tomato Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Tomato
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

transplanting, thrips, mites, fruit rot and virus. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Chilli is mainly associated with Kharif/Rabi. Broad planting reference: Region-specific. Disease examples: Leaf curl, powdery mildew, anthracnose. Pest examples: Thrips, mites, aphids, fruit borer. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruiting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Chilli outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Chilli Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Chilli
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

shoot and fruit borer, wilt and field sanitation. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CROP PROFILE

Brinjal is mainly associated with Multiple. Broad planting reference: Region-specific. Disease examples: Bacterial wilt, Phomopsis blight. Pest examples: Shoot and fruit borer, jassids, whitefly. Exact dates and management depend on local conditions.

GROWTH STAGES

Land preparation → planting/establishment → vegetative growth → flowering → filling/fruitletting → maturity → harvest. Record the stage because pest, disease, heat and water risks change during the crop cycle.

OFF-SEASON RISK

Growing Brinjal outside the locally suitable period can create temperature, rainfall, irrigation, pest, disease, maturity and market risks. Check local crop calendars and current agricultural advice before planting.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Brinjal Production?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Brinjal
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruitletting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

how conventional crop knowledge connects with precision agriculture. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Traditional + Modern Farming: how conventional crop knowledge connects with precision agriculture. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Traditional + Modern Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

components, layout, crop suitability, filtration, pressure and maintenance. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Water is delivered near the root zone through mains, laterals and emitters. Record filter condition, pressure, emitter discharge, spacing, irrigation duration and soil moisture. Main risks are clogging, leakage, pressure variation and poor scheduling.

FARM USE

Drip Irrigation should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Drip Irrigation?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

types, suitability, scheduling and precautions. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Pressurized water is distributed through sprinklers. Record sprinkler type, spacing, pressure and irrigation duration. Wind can cause drift and uneven distribution. Check pipes, nozzles, filters and pressure regularly.

FARM USE

Sprinkler Irrigation should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Sprinkler Irrigation?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

orchards, vegetables, coverage and management. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Micro-Sprinklers: orchards, vegetables, coverage and management. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Micro-Sprinklers?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

fertilizer delivery through irrigation, benefits and precautions. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Soluble nutrients are supplied through irrigation. Record product, nutrient, concentration, compatibility, crop stage and irrigation event. Poor mixing or water quality can cause precipitation and clogging; use source-verified recommendations.

FARM USE

Fertigation should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Fertigation?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

controllers, valves, sensors, scheduling and fail-safe practices. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Controllers operate valves using schedules and/or sensor inputs. Record settings, sensor inputs, valve status, irrigation duration and failures. Include safe behavior for power or connectivity loss.

FARM USE

Automated Irrigation should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Automated Irrigation?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

sensor types, placement, calibration and irrigation decisions. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Sensors estimate soil water status and can support irrigation scheduling. Record sensor type, depth, location, calibration, timestamp and reading. Poor placement or calibration can lead to wrong irrigation decisions.

FARM USE

Soil-Moisture Sensors should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Soil-Moisture Sensors?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

temperature, rainfall, humidity, wind and field-level monitoring. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Field stations measure variables such as temperature, rainfall, humidity and wind. Record station ID, timestamp, units and calibration. These data support irrigation and weather-risk decisions.

FARM USE

Weather Stations should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Weather Stations?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

sensors, connectivity, dashboards and alerts. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Connected sensors send field information to gateways, databases and dashboards. Record device ID, sensor, unit, time, location, battery and connectivity. Use authentication, backups and safe fallback behavior.

FARM USE

IoT Smart Farming should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is IoT Smart Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

decision support, prediction and farm recommendations. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

AI can combine weather, soil, crop and image data for prediction and decision support. Record model version, data period, features, confidence and validation. AI should support—not replace—agronomic verification.

FARM USE

AI in Farming should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is AI in Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

yield prediction, disease prediction and resource optimization. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

ML can predict yield, disease, pests or resource demand from historical data. Avoid duplicates and train/test leakage. Store features, labels, model version, validation method, accuracy and failure cases.

FARM USE

Machine Learning should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Machine Learning?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

crop images, pest/disease detection and limitations. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Images can be used for disease detection, pest counting, weed recognition and grading. Record crop, stage, image conditions and labels. Lighting, variety and camera differences can reduce accuracy.

FARM USE

Computer Vision should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Computer Vision?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

mapping, monitoring, spraying and safety. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Drones can support crop scouting, mapping, stress assessment and permitted precision operations. Record flight date, field boundary, sensor and weather. Consider people, livestock, wind, rain and regulations.

FARM USE

Agricultural Drones should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Agricultural Drones?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

precision application, drift, weather and regulatory precautions. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Drone spraying can provide targeted application where permitted. Record target, approved product, field area and weather. Follow current labels and applicable rules; never invent application rates or flight settings.

FARM USE

Drone Spraying should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Drone Spraying?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

NDVI/thermal/RGB imaging and scouting. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Drone Crop Monitoring: NDVI/thermal/RGB imaging and scouting. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Drone Crop Monitoring?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

crop health, water stress and large-area monitoring. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Satellite imagery can monitor crop extent, vegetation and stress indicators. Record source, date, resolution, cloud cover and field boundary. Ground verification is needed because imagery can be affected by clouds, soil and crop stage.

FARM USE

Satellite Remote Sensing should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Satellite Remote Sensing?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

field maps, soil maps, irrigation zones and spatial analysis. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

GIS combines field boundaries with soil, crop, irrigation and pest layers. Record coordinate system, source date and field ID. Incorrect or outdated maps can produce incorrect decisions.

FARM USE

GIS in Agriculture should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is GIS in Agriculture?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

field boundaries, guidance and machine positioning. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

GPS supports field mapping, machinery guidance and repeatable operations. Record receiver, field boundary, accuracy and operation date. Check boundaries and signal quality before machinery work.

FARM USE

GPS Precision Farming should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is GPS Precision Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

site-specific fertilizer, seed and input application. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Inputs can be changed according to mapped field variation. Record prescription map, equipment settings and actual application. Poor maps or calibration can reduce benefits.

FARM USE

Variable Rate Technology should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

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FARMER / RAG QUESTIONS

- What is Variable Rate Technology?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

seed placement, spacing, depth and machine control. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Precision planters control seed spacing, depth and placement. Record seed lot, planter settings, spacing, depth and field condition. Calibration and seed quality remain important.

FARM USE

Precision Seeding should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Precision Seeding?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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Action:	Source-backed management
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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

GPS guidance, automation, telemetry and data. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Smart tractors can combine GPS, sensors, telemetry and implement data. Record machine, operation, field, settings and maintenance. Operator training and machine safety remain necessary.

FARM USE

Smart Tractors should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Smart Tractors?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
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PURPOSE

weeding, harvesting, monitoring and automation. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Agricultural Robotics: weeding, harvesting, monitoring and automation. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Agricultural Robotics?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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Action:	Source-backed management
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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

mechanical/robotic systems and crop recognition. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Automated Weeding: mechanical/robotic systems and crop recognition. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Automated Weeding?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

protected climate, sensors and controlled production. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Protected structures allow closer control of temperature, humidity, light, ventilation and irrigation. Record crop stage, climate readings, irrigation and pests. Excess heat or humidity can create serious problems.

FARM USE

Greenhouse Farming should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Greenhouse Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

environment control, irrigation, fertigation and crop protection. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Polyhouses provide protected crop conditions for selected vegetables, flowers and nurseries. Record ventilation, temperature, humidity, irrigation and crop stage. Pest entry, high humidity and structural maintenance need attention.

FARM USE

Polyhouse Farming should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Polyhouse Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

light/temperature moderation and crop selection. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Shade-Net Farming: light/temperature moderation and crop selection. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Shade-Net Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

soilless systems, lighting, nutrients and energy. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Crops are grown in stacked controlled environments. Record light, temperature, humidity, nutrient solution and energy use. Energy cost and climate control are major considerations.

FARM USE

Vertical Farming should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Vertical Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

nutrient solution, pH, EC and root-zone management. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Plants grow without field soil using nutrient solution. Record pH, EC, water temperature, nutrient solution, flow and root condition. Root disease, nutrient imbalance and power failure are important risks.

FARM USE

Hydroponics should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Hydroponics?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

root misting, control requirements and risks. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Roots receive nutrient mist rather than soil. Record misting cycle, pressure, nutrient solution and root condition. Nozzle blockage or power failure can quickly stress plants.

FARM USE

Aeroponics should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Aeroponics?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

fish-plant integration and water quality. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Aquaponics: fish-plant integration and water quality. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Aquaponics?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

solar pumps, storage, controllers and water planning. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Solar panels can power irrigation pumps. Record solar capacity, pump capacity, operating time and water source. Easy daytime pumping should not lead to unnecessary water extraction.

FARM USE

Solar Irrigation should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Solar Irrigation?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

solar, biogas and energy-efficient systems. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Renewable Energy on Farms: solar, biogas and energy-efficient systems. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Renewable Energy on Farms?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

pheromone, sticky, camera and sensor-based monitoring. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Sensors, cameras or specialized traps can monitor pest populations. Record trap type, lure, location, crop stage, count and date. Poor placement or wrong lure can produce misleading results.

FARM USE

Smart Pest Traps should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Smart Pest Traps?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

image-based identification, confidence and expert verification. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

AI Disease Detection: image-based identification, confidence and expert verification. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is AI Disease Detection?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

weather, soil, crop-stage and historical data. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

AI Yield Prediction: weather, soil, crop-stage and historical data. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is AI Yield Prediction?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

mobile alerts, localized recommendations and records. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Mobile or web systems can deliver crop, weather and pest information. Record farmer location, crop, date, source and advisory version. Advice must be current and locally relevant.

FARM USE

Digital Farm Advisory should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Digital Farm Advisory?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

crop calendars, weather, market and advisory features. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Mobile Farmer Applications: crop calendars, weather, market and advisory features. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Mobile Farmer Applications?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

field records, inputs, labor, costs and yield. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Digital records can track fields, seed, fertilizer, labor, machinery, costs and harvest. Use stable field IDs and units. Backups and data validation reduce errors.

FARM USE

Farm Management Software should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Farm Management Software?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

pricing, buyers, traceability and logistics. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Digital Marketplaces: pricing, buyers, traceability and logistics. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Digital Marketplaces?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
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Observation:	Symptom, pest count, weather or technology reading
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Action:	Source-backed management
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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

farm-to-market provenance and limitations. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Blockchain Traceability: farm-to-market provenance and limitations. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Blockchain Traceability?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

temperature, humidity, sensors and pest monitoring. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Sensors can monitor temperature and humidity in storage. Record lot, commodity, temperature, humidity and inspection result. Condensation, pests and sensor failure can reduce quality.

FARM USE

Smart Storage should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Smart Storage?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

pre-cooling, transport, storage and quality preservation. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Temperature-controlled storage and transport protect perishable products. Record lot, temperature, time and transport events. Temperature breaks and delays should be traceable.

FARM USE

Cold Chain should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Cold Chain?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

sorting, grading, packing and quality inspection. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Post-Harvest Automation: sorting, grading, packing and quality inspection. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Post-Harvest Automation?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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Action:	Source-backed management
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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

combining satellite and field observations. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Remote Sensing + Weather: combining satellite and field observations. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Remote Sensing + Weather?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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Action:	Source-backed management
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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

soil maps, sensors and variable application. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Precision Nutrient Management: soil maps, sensors and variable application. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Precision Nutrient Management?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

crop demand, sensors and weather forecasts. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Precision Water Management: crop demand, sensors and weather forecasts. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Precision Water Management?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

technology for drought, heat, flood and variability. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Climate-smart practices improve resilience and resource efficiency through water conservation, diversification, soil protection and suitable technology. Always adapt the approach to local climate and farm resources.

FARM USE

Climate-Smart Farming should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Climate-Smart Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

biological inputs plus digital monitoring. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Modern Organic Farming: biological inputs plus digital monitoring. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Modern Organic Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

microbial inputs, quality and application records. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Beneficial microorganisms can support nutrient availability or plant growth. Record organism/product, crop, application and storage. Product quality and viability matter.

FARM USE

Biofertilizers should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Biofertilizers?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

microbial/botanical options and IPM integration. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Biological or naturally derived pest-management products can be part of IPM. Record target pest, crop, product and timing. Storage, quality and environmental conditions influence performance.

FARM USE

Biopesticides should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Biopesticides?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

sensor networks and disease scouting. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Protected Crop Monitoring: sensor networks and disease scouting. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Protected Crop Monitoring?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

sensor → gateway → cloud → analytics → actuator. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Farm Automation Architecture: sensor → gateway → cloud → analytics → actuator. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Farm Automation Architecture?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

local decisions when internet connectivity is weak. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Edge Computing in Farming: local decisions when internet connectivity is weak. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Edge Computing in Farming?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

data storage, dashboards and analytics. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Cloud Agriculture Platforms: data storage, dashboards and analytics. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Cloud Agriculture Platforms?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

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- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

virtual field representation and scenario analysis. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Digital Twins for Farms: virtual field representation and scenario analysis. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Digital Twins for Farms?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
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- Use integrated pest management and current source-verified recommendations.
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PURPOSE

accounts, device security, backups and privacy. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

TECHNOLOGY

Digital farms need secure accounts, device protection, backups and controlled access. Record ownership and permissions. Weak passwords, malware and lost devices can cause data loss.

FARM USE

Farm Data Security should be adopted for a measurable objective such as water saving, early detection, input efficiency, labor reduction, quality improvement or better decision support. Record baseline and outcome.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Farm Data Security?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

standard field names, units and data exchange. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Interoperability: standard field names, units and data exchange. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Interoperability?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

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- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
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PURPOSE

capital cost, operating cost, labor savings and ROI. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Modern Tech Economics: capital cost, operating cost, labor savings and ROI. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

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MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Modern Tech Economics?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

EXAMPLE DATA RECORD

Crop:	Exact crop
Season:	Kharif / Rabi / Zaid / Perennial
Region:	State + District
Growth stage:	Vegetative / Flowering / Fruiting / Maturity
Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
Source:	ICAR / IMD / KVK / University / Government

IMPORTANT NOTES

- Crop calendars are local; a season label alone is not a universal sowing date.
- Record symptoms together with crop stage, weather, affected plant part and field history.
- Prevention includes healthy planting material, sanitation, suitable timing, balanced nutrition, water management and scouting.
- Use integrated pest management and current source-verified recommendations.
- Technology is decision support: sensors, AI and drones can fail and require validation.
- Every important dataset record should preserve source, date and version.

PURPOSE

skills, training, service models and smallholder access. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Technology Adoption: skills, training, service models and smallholder access. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

Kharif is broadly monsoon-linked, Rabi is broadly winter-linked and Zaid is generally the summer/inter-season period. Exact dates vary by district. Record rainfall, temperature, humidity and wind because these affect crop growth, irrigation and disease/pest risk.

PRECAUTIONS

Use healthy planting material, suitable varieties, appropriate planting time, sanitation, balanced nutrition, weed control, appropriate irrigation and drainage, rotation where suitable and regular scouting. For chemical products, verify the current official label and local recommendation.

MODERN FARMING CONNECTION

Drip irrigation, fertigation, soil sensors, automated irrigation, weather stations, IoT, drones, satellite imagery, GIS/GPS, precision seeding, variable-rate technology, AI/ML, protected cultivation, hydroponics and robotics can improve farming when correctly designed and maintained.

DATASET FIELDS

Recommended fields: record_id, crop, variety, region, season, date, growth_stage, soil, irrigation, weather, pest, disease, symptoms, incidence, severity, risk, precaution, action, technology, source, publication_date, version, reviewer and outcome.

FARMER / RAG QUESTIONS

- What is Technology Adoption?
- What season or growth stage is relevant?
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PURPOSE

shared access to drones and machinery. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Custom Hiring Centres: shared access to drones and machinery. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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FARMER / RAG QUESTIONS

- What is Custom Hiring Centres?
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- What precautions should be taken?
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Observation:	Symptom, pest count, weather or technology reading
Precaution:	Source-backed preventive measure
Action:	Source-backed management
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PURPOSE

calibration, cleaning, battery and replacement. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Sensor Maintenance: calibration, cleaning, battery and replacement. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

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FARMER / RAG QUESTIONS

- What is Sensor Maintenance?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
- What source supports the recommendation?

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PURPOSE

filters, flushing, pressure and emitter checks. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Drip Maintenance: filters, flushing, pressure and emitter checks. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

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FARMER / RAG QUESTIONS

- What is Drip Maintenance?
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EXAMPLE DATA RECORD

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PURPOSE

sensor errors, clogging, pressure and connectivity. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Smart Irrigation Troubleshooting: sensor errors, clogging, pressure and connectivity. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

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FARMER / RAG QUESTIONS

- What is Smart Irrigation Troubleshooting?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
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PURPOSE

operator training, weather, people, livestock and chemical safety. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Drone Safety: operator training, weather, people, livestock and chemical safety. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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PRECAUTIONS

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MODERN FARMING CONNECTION

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DATASET FIELDS

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FARMER / RAG QUESTIONS

- What is Drone Safety?
- What season or growth stage is relevant?
- What are the major risks?
- What symptoms or warning signs should be recorded?
- What precautions should be taken?
- How can modern technology help?
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PURPOSE

false alerts, sensor errors, cost and connectivity. This page is part of the 100-page farming dataset and contains practical crop, risk, precaution and technology information.

CORE EXPLANATION

Technology Limitations: false alerts, sensor errors, cost and connectivity. Connect the topic to crop, season, growth stage, soil, water, weather, pest/disease pressure and farm economics.

SEASON + WEATHER

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FARMER / RAG QUESTIONS

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